

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	SWTID-2026-5841
Project Title	FitBuddy – AI Fitness Plan Generator
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

FitBuddy uses Generative AI to give people a personalised 7-day workout plan and a nutrition tip in seconds, and lets them improve the plan with plain-language feedback. It addresses the lack of affordable, adjustable fitness guidance and aims for safe, simple, general wellness support.

Project Overview

Objective	To build a web application that generates a safe, personalised weekly workout plan from a user's age, weight, goal and intensity, and revises it from feedback.
Scope	Web form, AI plan generation, nutrition tip, feedback-based plan update, storage of users and plans, and an admin page to view or delete users. Intended for general wellness; it does not give medical advice.

Problem Statement

Description	Beginners and casual exercisers do not know what to do each day, generic plans ignore their details, and personal trainers are costly, so many people quit early.
Impact	A personalised, adjustable plan available to everyone increases the chance that people start and continue exercising, improving wellness at very low cost.

Proposed Solution

Approach	Send the validated user profile to Google Gemini with a safety-focused prompt, store the result, and let the user submit feedback so that a complete revised plan is generated from the latest version.
Key Features	- Personalised 7-day plan with warm-up, main workout and cooldown - Short nutrition / recovery tip for the chosen goal - Feedback-based plan revision - Saved user records and admin dashboard - Retry and error handling for AI failures

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications, number of cores	Any modern 4-core CPU; no GPU needed (AI runs on Google's servers)
Memory	RAM specifications	8 GB

Resource Type	Description	Specification / Allocation
Storage	Disk space for code, environment and database	1 GB free space
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries	Additional libraries	SQLAlchemy, Pydantic, python-multipart, google-genai, python-dotenv, pytest, httpx
Development Environment	IDE and tools	Python 3.12, code editor / terminal, Git and GitHub, Docker (optional)
Data		
Data	Source, size, format	User-entered form data and AI-generated text; one record per user; stored in an SQLite table (users)